BATCH NO:MI1234

PERSONALIZED LEARNING ROADMAP AND RESOURCES RECOMMENDATION SYSTEM

Minor Project-I report submitted in partial fulfillment of the requirement for award of the degree of

Bachelor of Technology in Computer Science & Engineering

By

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Under the guidance of
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SCHOOL OF COMPUTING

VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF SCIENCE AND TECHNOLOGY
(Deemed to be University)
Accredited by NAAC with A++ Grade
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CERTIFICATE

It is certified that the work contained in the project report titled ”PERSONALIZED LEARNING ROADMAP AND RESOURCES RECOMMENDATION SYSTEM” by ”M. HARINATH REDDY (24UECS0732), M. HIMA TEJA (24UECS0737), C. HIMATEJA (24UECS1447)” has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

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Institute of Science and Technology

DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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APPROVAL SHEET

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ABSTRACT

The Personalized Learning Roadmap and Resources Recommendation System is a web-based platform designed to help students plan and manage their learning according to their individual skills, interests, and career goals. The system collects student information and performs skill-gap analysis to identify existing skills and the skills required for the selected career path. Based on this analysis, it generates a personalized step-by-step learning roadmap and recommends relevant learning resources such as courses, videos, books, tutorials, documentation, and practice materials. The system also provides career alignment by connecting learning activities with the skills required for the desired career role. A progress-tracking module allows students to monitor completed topics and remaining skills. By integrating these features into a single platform, the proposed system provides an organized and personalized learning experience. In the future, AI-based recommendation techniques can be incorporated to make the roadmap more adaptive based on student performance and learning behavior.

Keywords:

Personalized Learning, Learning Roadmap, Resource Recommendation, Student Profiling, Skill-Gap Analysis, Career Alignment, Progress Tracking, Artificial Intelligence, Machine Learning, E-Learning.

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LIST OF ACRONYMS AND ABBREVIATIONS

- **AI** – Artificial Intelligence
- **ML** – Machine Learning
- **UI** – User Interface
- **UX** – User Experience
- **DFD** – Data Flow Diagram
- **UML** – Unified Modeling Language
- **DB** – Database
- **SQL** – Structured Query Language
- **HTML** – HyperText Markup Language
- **CSS** – Cascading Style Sheets
- **JS** – JavaScript
- **API** – Application Programming Interface
- **NLP** – Natural Language Processing
- **IDE** – Integrated Development Environment
- **CRUD** – Create, Read, Update, Delete

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Chapter 1

INTRODUCTION

1.1 Introduction

In today's digital learning environment, students have access to a large number of online courses, videos, books, tutorials, and practice platforms. However, choosing the right resources and deciding what to learn, in what order, and according to which career goal can be difficult. Students with different skill levels and interests often follow the same learning paths, which may lead to confusion, unnecessary learning, and difficulty in identifying required skills.

The Personalized Learning Roadmap and Resources Recommendation System is proposed to address these challenges by providing a learning path based on each student's individual requirements. The system collects information such as existing skills, interests, strengths, weaknesses, learning preferences, and career goals. It then performs skill-gap analysis to identify the skills that need to be developed.

Based on the identified skill gaps, the system generates a personalized step-by-step learning roadmap and recommends suitable resources such as courses, videos, books, tutorials, documentation, and practice materials. The system also aligns the roadmap with the student's desired career path and provides progress tracking to monitor completed and pending learning activities.

The proposed system integrates student profiling, skill-gap analysis, personalized roadmap generation, resource recommendation, career alignment, and progress tracking into a single platform. This approach aims to make the learning process more structured, flexible, and relevant to individual career objectives. In future development, AI and machine learning techniques can be incorporated to provide more adaptive recommendations based on student performance and learning behavior.

1.2 Background

The rapid growth of online education has provided students with access to a wide range of learning resources. Platforms such as online courses, video tutorials, e-books, documentation, and coding websites offer valuable opportunities for skill development. However, the large amount of available content can make it difficult for students to identify which skills to learn, which resources to use, and in what sequence they should learn them.

Students have different educational backgrounds, skill levels, interests, learning abilities, and career goals. A common learning path may not be suitable for every student. Some students may spend time learning skills they already know, while others may miss important skills required for their desired career.

The Personalized Learning Roadmap and Resources Recommendation System aims to address this issue by analyzing the student's current profile and career objectives. It identifies skill gaps and generates a structured learning roadmap with relevant resources. The system also provides progress tracking to help students monitor their development.

By bringing these functionalities together in a single platform, the proposed system can provide a more organized and personalized learning experience. The system can also be enhanced in the future using Artificial Intelligence and Machine Learning to improve recommendations based on individual learning progress and performance.

1.3 Objective

The primary objective of the Personalized Learning Roadmap and Resources Recommendation System is to develop a platform that helps students create a structured and personalized learning path based on their current skills, interests, strengths, weaknesses, and career goals. The system aims to identify the gap between the student's existing knowledge and the skills required for their desired career. Based on this analysis, it generates a step-by-step learning roadmap that helps students understand what they should learn first, what they should learn next, and which skills are essential for achieving their career objectives. The system also aims to recommend suitable learning resources such as online courses, videos, books, tutorials, documentation, and practice materials.

Another important objective is to provide an integrated platform where students can manage their learning journey and monitor their progress in one place. The system aims to align recommended learning activities with specific career roles and provide progress tracking for completed and pending tasks. It also seeks to reduce confusion caused by the large amount of available online learning content by presenting relevant resources in an organized manner. The proposed system can be further enhanced using Artificial Intelligence and Machine Learning techniques to provide adaptive recommendations based on student performance, learning behavior, and progress. Overall, the objective is to make learning more personalized, organized, career-oriented, and efficient.

1.4 Problem Statement

The rapid growth of online education has made a large number of learning resources available to students, including online courses, video lectures, books, tutorials, documentation, and practice platforms. Although this availability provides more learning opportunities, it also creates a major problem: students often struggle to identify what they need to learn, where to start, which resources are relevant, and how to organize their learning according to their career goals. Most existing learning platforms provide general recommendations or predefined learning paths that may not consider an individual student's current skills, interests, strengths, weaknesses, and career objectives. As a result, students may spend time learning unnecessary topics, miss important skills, or follow an unsuitable learning sequence.

Another challenge is the lack of an integrated system that combines skill-gap identification, personalized roadmap generation, resource recommendation, career alignment, and progress tracking in one platform. Students often need to use multiple websites to find courses, track their progress, and understand the skills required for a particular career. This can make the learning process unorganized and difficult to manage. Therefore, there is a need for a system that analyzes a student's existing skills and desired career path, identifies the required skill gaps, creates a personalized step-by-step learning roadmap, and recommends suitable learning resources. The proposed Personalized Learning Roadmap and Resources Recommendation System aims to address these problems by providing students with an organized and personalized approach to achieving their learning and career objectives.

Chapter 2

LITERATURE REVIEW

Existing studies on personalized learning systems focus on recommending educational content based on students' interests, skills, and learning behavior. Recommendation techniques such as content-based filtering and collaborative filtering are commonly used to suggest suitable courses, videos, and study materials.

Research also highlights skill-gap analysis for identifying the difference between a student's existing skills and the skills required for a specific career. AI and Machine Learning can further improve recommendations by considering student performance and learning progress.

However, many existing systems focus mainly on resource recommendation and do not combine skill-gap analysis, personalized roadmaps, career alignment, resource recommendation, and progress tracking in a single platform. The proposed system aims to integrate these features to provide students with a structured and personalized learning experience.

2.1 Existing System

Currently, students use different online learning platforms such as YouTube, Coursera, Udemy, and educational websites to learn new skills. These platforms provide a wide range of courses and learning materials, but the recommendations are generally based on popularity, ratings, searches, or previously viewed content. They may not fully consider the student's current skills, weaknesses, career goals, and required skill gaps.

2.2 Related Work

Existing e-learning systems provide courses and learning resources based on students' interests and activities. Some recommendation systems use content-based filtering, collaborative filtering, AI, and Machine Learning to suggest suitable resources.

Skill-gap analysis systems help identify missing skills for specific careers. However, most existing systems do not combine skill analysis, personalized roadmaps, resource recommendations, career alignment, and progress tracking in one platform. The proposed system integrates these features to provide a complete personalized learning experience.

2.3 Research Gap

Existing learning platforms provide many educational resources, but they often do not provide a complete personalized learning path based on a student's current skills and career goals. Most systems focus mainly on course or resource recommendations and provide limited support for identifying skill gaps and organizing learning activities in the correct sequence.

There is also a lack of a single integrated platform that combines student profiling, skill-gap analysis, personalized roadmap generation, resource recommendation, career alignment, and progress tracking. The proposed system addresses this gap by integrating these features into one platform, helping students follow a structured learning path according to their individual requirements and career objectives.

Chapter 3

PROJECT DESCRIPTION

3.1 Existing System

Currently, students depend on various online learning platforms and educational websites to develop their technical and professional skills. Platforms such as online course websites, video-learning platforms, coding websites, and digital libraries provide a large collection of courses, tutorials, videos, books, and practice materials. However, these platforms generally provide recommendations based on search history, popularity, ratings, or previously viewed content. They may not consider the student's complete profile, including current skills, strengths, weaknesses, interests, preferred learning methods, and specific career goals. Students therefore need to manually search for suitable resources and decide which topics to learn and in what order.

Another limitation is that existing systems usually do not integrate skill-gap analysis, personalized roadmap generation, career alignment, resource recommendation, and progress tracking into a single platform. Students may use different applications for learning resources, career guidance, and progress management. This can make the learning process difficult to organize and may result in students following unsuitable or repetitive learning paths.

Disadvantages of existing methods:

1. Limited personalization for individual students.
2. No proper skill-gap analysis.
3. Students need to create their own learning roadmap.
4. Resources are scattered across different platforms.
5. Limited career-based recommendations.
6. Progress tracking is not integrated.

3.2 Proposed System

The proposed Personalized Learning Roadmap and Resources Recommendation System is designed to provide students with a structured and personalized learning experience based on their individual skills, interests, strengths, weaknesses, and career goals. The system first collects the student's profile information and identifies their existing skills. It then performs skill-gap analysis by comparing the student's current skills with the skills required for the selected career path. Based on the identified gaps, the system generates a step-by-step learning roadmap that guides the student from their current level towards their desired career goal.

The system also recommends relevant courses, videos, books, tutorials, documentation, and practice materials for each required skill. It connects the learning roadmap with the student's career objective and provides a progress-tracking feature to monitor completed and pending activities. All these functions are integrated into a single platform, reducing the need to use multiple systems. In the future, AI and Machine Learning techniques can be incorporated to provide more adaptive recommendations based on student performance and learning behavior.

Advantages of Proposed System

1. Provides personalized learning paths.
2. Identifies individual skill gaps.
3. Recommends relevant learning resources.
4. Aligns learning with career goals.
5. Tracks student learning progress.
6. Integrates multiple features into one platform.
7. Can be enhanced with AI-based recommendations.

3.3 Feasibility Study

The Personalized Learning Roadmap and Resources Recommendation System is feasible to develop as a web-based minor project using commonly available technologies and resources. From a technical feasibility perspective, the system can be developed using technologies such as HTML, CSS, JavaScript, Python, Flask, and MySQL. Student profiles, skills, career requirements, learning roadmaps, recommended resources, and progress information can be stored in a database. The recommendation process can initially use skill matching and rule-based techniques, while AI or Machine Learning can be added as a future enhancement. From an economic feasibility perspective, the project can be developed using open-source software and freely available development tools, reducing development costs. From an operational feasibility perspective, the system provides a simple interface through which students can enter their skills and career goals, view their personalized roadmap, access recommended resources, and track their progress. The system does not require complex hardware and can run on commonly available computers with an internet connection. Therefore, the proposed system is technically, economically, and operationally feasible for implementation as a minor project.

3.3.1 Economic Feasibility

The Personalized Learning Roadmap and Resources Recommendation System is economically feasible because it can be developed using mostly free and open-source technologies. The development team can use tools such as HTML, CSS, JavaScript, Python, Flask, and MySQL without requiring expensive software licenses. The system can be developed and tested on existing computers, which reduces the need for additional hardware investment. A local database can be used during development, avoiding the cost of purchasing a dedicated server. Similarly, freely available learning resources and publicly accessible educational content can be included in the recommendation database. Since the project is intended as a minor project, the initial development and maintenance requirements are relatively low. If the system is deployed online, a low-cost hosting service can be used based on the number of users and storage requirements. The system can also reduce the effort required by students to search for suitable learning paths and resources by providing recommendations in one platform.

3.3.2 Technical Feasibility

The Personalized Learning Roadmap and Resources Recommendation System is technically feasible because it can be developed using commonly available software, programming languages, and database technologies. The system can use HTML, CSS, and JavaScript for developing the user interface and Python with Flask for implementing the backend logic. MySQL can be used to store student profiles, skills, career requirements, learning resources, roadmaps, and progress information. The skill-gap analysis can be implemented using predefined skill mappings and matching techniques to compare the student's existing skills with the skills required for the selected career. A recommendation module can then suggest relevant learning resources based on the identified skill gaps. The roadmap generation module can arrange the required skills in a logical learning sequence. The system can also provide a dashboard for viewing roadmaps and tracking completed activities. Since these technologies are widely available and can run on standard computers, no specialized hardware is required. AI and Machine Learning techniques can be integrated later to improve recommendation accuracy. Therefore, the proposed system is technically practical and suitable for implementation as a minor project.

3.3.3 Social Feasibility

The Personalized Learning Roadmap and Resources Recommendation System is socially feasible because it is designed to support students in making their learning process more organized, accessible, and career-oriented. Students often face difficulty in selecting suitable learning resources and deciding which skills they should develop for their desired career. The proposed system helps address these difficulties by providing a personalized roadmap based on the student's existing skills, interests, and career goals. By presenting relevant learning resources and a clear sequence of topics, the system can reduce confusion and help students plan their learning more effectively. The progress-tracking feature also allows students to monitor their completed activities and remaining learning tasks, which can encourage consistent learning. The system can be accessed through a simple web interface and does not require specialized technical knowledge to use. It can support students from different skill levels by providing recommendations according to their individual requirements.

3.4 System Specification

Hardware Requirements:

- **Processor:** Intel Core i5 / AMD Ryzen 5 or higher
- **RAM:** Minimum 8 GB; 16 GB recommended
- **Storage:** Minimum 256 GB SSD; 512 GB SSD recommended
- **Display:** Full HD (1920×1080) or higher
- **Internet:** Stable broadband/Wi-Fi connection
- **Graphics:** Integrated graphics sufficient
- **Input Devices:** Keyboard and mouse

Software Requirements:

- **Operating System:** Windows 11 / Ubuntu 22.04 or later
- **Frontend:** HTML5, CSS3, JavaScript
- **Backend:** Python 3.11 or later
- **Framework:** Flask 3.x
- **Database:** MySQL 8.0 or later
- **Development Environment:** Visual Studio Code
- **Web Browser:** Google Chrome / Microsoft Edge (latest version)
- **Version Control:** Git and GitHub
- **API Testing:** Postman
- **Optional AI/ML:** Python libraries such as Scikit-learn, Pandas, and NumPy
- **Deployment:** Render / Railway / similar cloud platform

3.4.1 Tools and Technologies Used

HTML – Web page structure.
CSS – User interface design.
JavaScript – Interactive features.
Python – Backend development.
Flask – Web application framework.
MySQL – Database management.
VS Code – Code development.
Git GitHub – Version control.
Pandas NumPy – Data processing.
Scikit-learn – Recommendation and skill matching.
Chrome – Testing the web application.

3.4.2 Standards and Policies

The Personalized Learning Roadmap and Resources Recommendation System follows basic software, security, and data-management standards to ensure that the system is reliable, secure, and easy to use.

Web Standards – Use HTML, CSS, and JavaScript standards.
Data Privacy – Protect student information.
Security – Secure user accounts and passwords.
Database Standards – Store and manage data properly.
Accessibility – Keep the system simple and user-friendly.
Copyright Policy – Use learning resources legally.
Testing Standards – Test the system for accuracy and performance.

Chapter 4

SYSTEM DESIGN AND METHODOLOGY

4.1 System Architecture

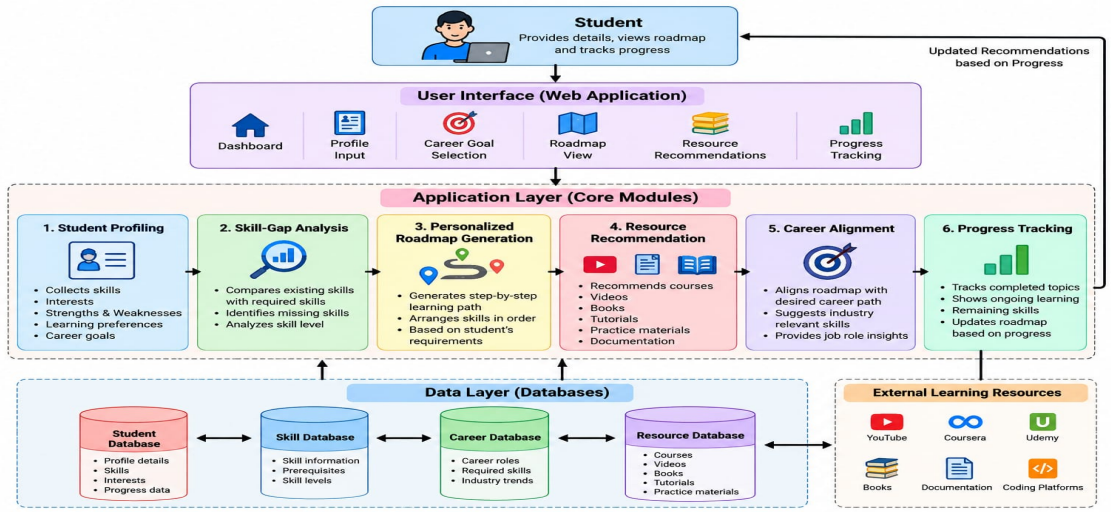


Figure 4.1: **Proposed System Architecture**

Description

The proposed system consists of four main parts: User Interface, Application Layer, Data Layer, and External Learning Resources. The User Interface allows students to enter their skills, interests, and career goals and view their personalized roadmap, recommended resources, and progress. The Application Layer performs student profiling, skill-gap analysis, roadmap generation, resource recommendation, career alignment, and progress tracking. The Data Layer stores student details, skills, career information, learning resources, and progress data. The system can also use external learning resources such as YouTube, Coursera, Udemy, books, documentation, and coding platforms. Based on the student's information, the system identifies missing skills, creates a personalized learning roadmap, and recommends suitable resources. The student's progress is continuously updated to provide an organized and personalized learning experience.

4.2 Design Phase

The Design Phase defines how the Personalized Learning Roadmap and Resources Recommendation System will work. The system is designed with modules for student profiling, skill-gap analysis, personalized roadmap generation, resource recommendation, career alignment, and progress tracking. Student details and learning information are stored in a database, while the application processes the data to generate suitable learning paths and resources. The design focuses on providing a simple, user-friendly interface and an organized learning experience.

4.2.1 Data Flow Diagram

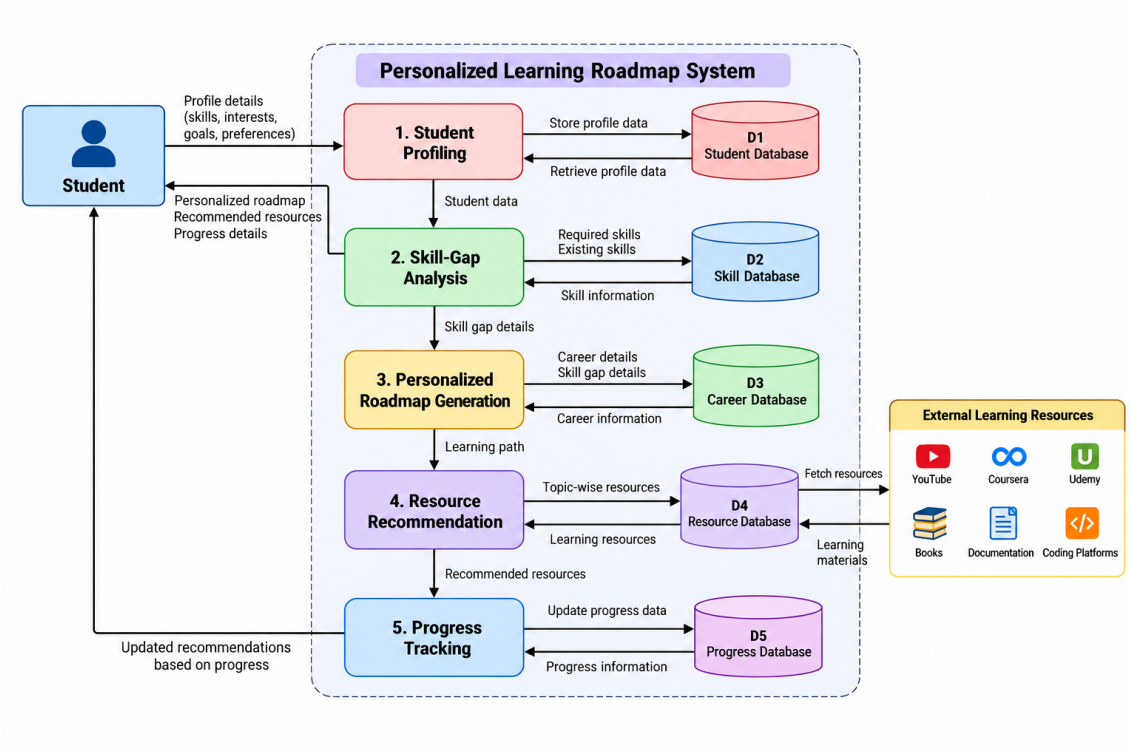


Figure 4.2: **Data Flow Diagram**

Description

The DFD shows how information moves through the system. The student provides profile details, skills, interests, and career goals. The system analyzes the skill gap and identifies the required skills. Based on this information, it creates a personalized learning roadmap and recommends suitable learning resources such as courses, videos, books, and tutorials. The system stores student, skill, career, resource, and

progress data in the database. Finally, the progress tracking module records completed activities and updates recommendations based on the student's progress.

4.2.2 Use Case Diagram

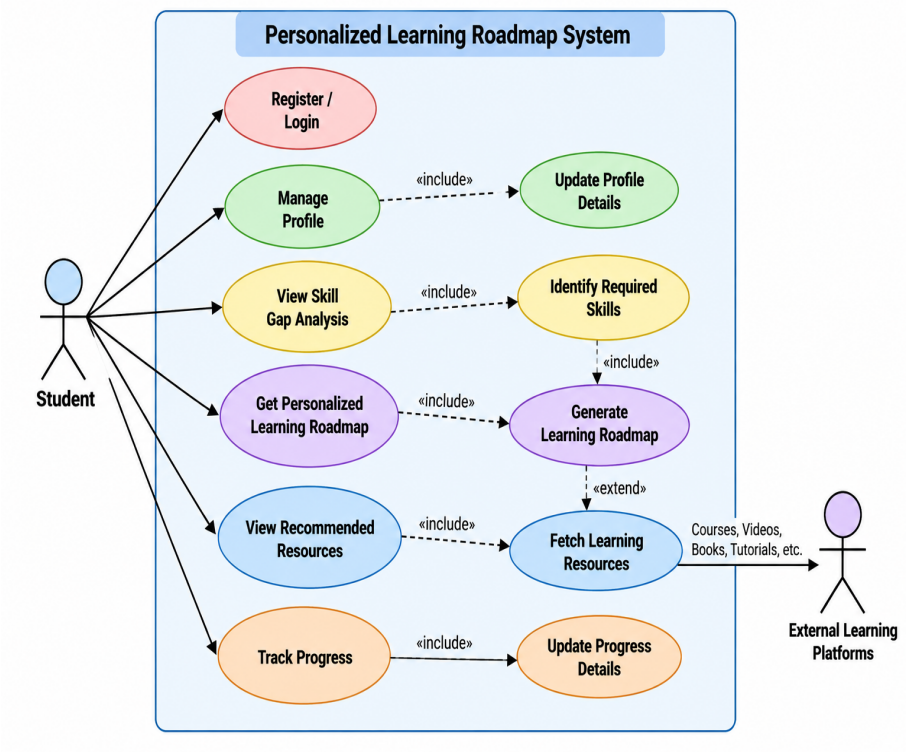


Figure 4.3: **Use Case Diagram**

Description

The Use Case Diagram shows how the student interacts with the Personalized Learning Roadmap System. The student can register/login, manage their profile, view skill-gap analysis, get a personalized learning roadmap, view recommended resources, and track progress. The system analyzes the student's skills and identifies the skills required for the selected career. It then generates a suitable learning roadmap and recommends resources such as courses, videos, books, and tutorials. The student can also update their progress, which helps the system provide updated recommendations. External learning platforms provide the required learning resources to the system.

4.2.3 Class Diagram

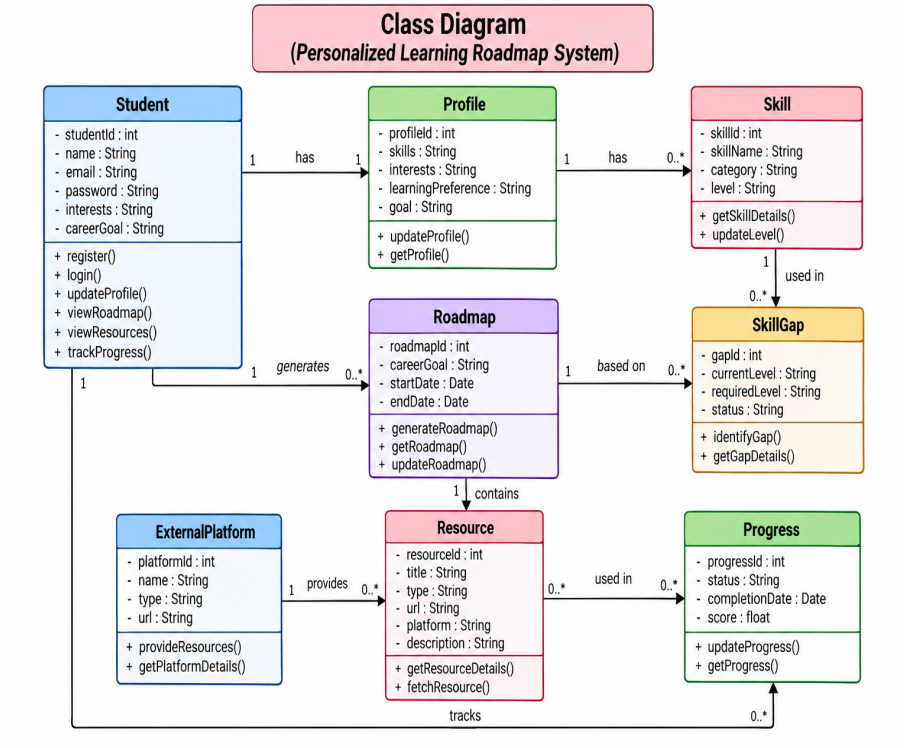


Figure 4.4: **Class Diagram**

Description

The Class Diagram shows the main classes and their relationships in the system. The Student class manages student details and interacts with the Profile class, which stores skills, interests, learning preferences, and career goals. The Skill and Skill-Gap classes identify existing and required skills. The Roadmap class generates and manages the student's personalized learning path.

The Resource class stores recommended learning materials such as courses, videos, books, and tutorials, while the ExternalPlatform class represents platforms that provide these resources. The Progress class records completed activities and learning progress. Together, these classes support student profiling, skill-gap analysis, roadmap generation, resource recommendation, and progress tracking.

4.2.4 Sequence Diagram

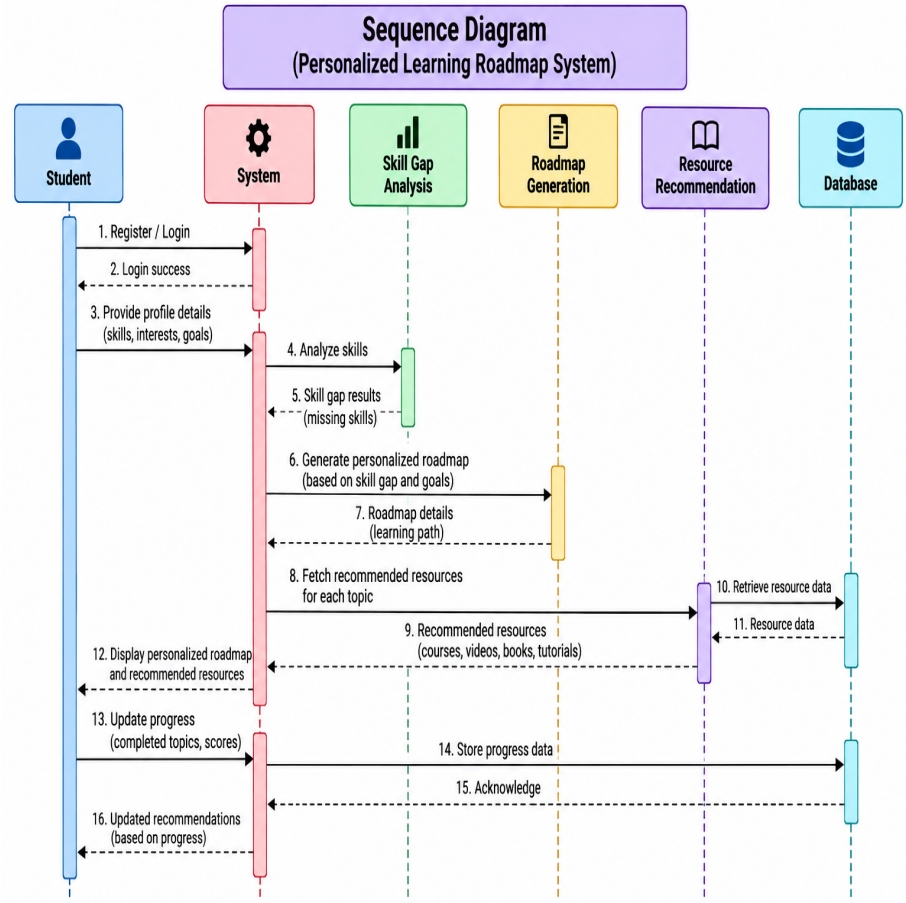


Figure 4.5: **Sequence Diagram**

Description

The Sequence Diagram shows the order of interactions between the student and the system. First, the student registers/logs in and provides profile details such as skills, interests, and career goals. The system performs skill-gap analysis to identify missing skills. Based on the results, it generates a personalized learning roadmap.

Next, the system retrieves suitable learning resources such as courses, videos, books, and tutorials from the database and displays them to the student. The student then updates their learning progress, and the system stores the progress data. Based on the updated progress, the system can provide updated recommendations to support the student's learning journey.

4.2.5 Collaboration diagram

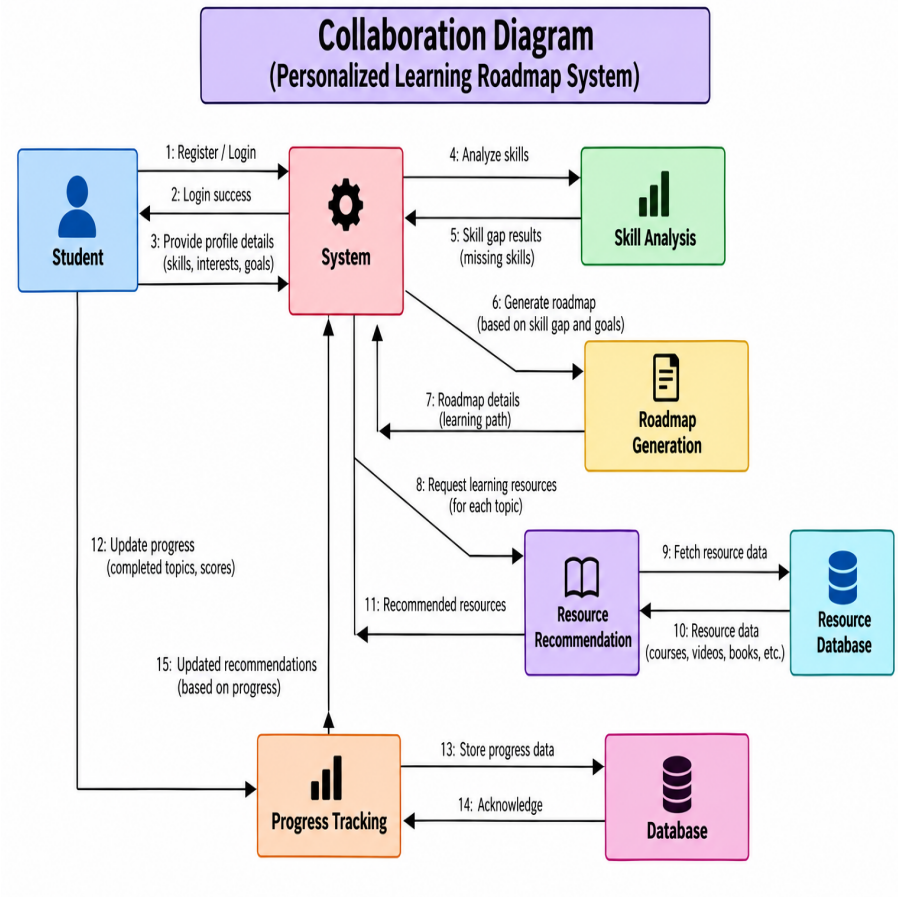


Figure 4.6: **Collaboration diagram**

Description

The Collaboration Diagram shows how different components of the system communicate with each other. The Student interacts with the System by logging in and providing profile details. The System sends the information to the Skill Analysis module to identify missing skills. The results are then passed to the Roadmap Generation module to create a personalized learning path.

The System also communicates with the Resource Recommendation module to obtain suitable courses, videos, books, and tutorials from the Resource Database. The Progress Tracking module records completed activities in the database. Based on the student's progress, the system provides updated learning recommendations. Thus, the diagram shows the communication and data exchange between the student,

system modules, and databases.

4.2.6 Activity Diagram

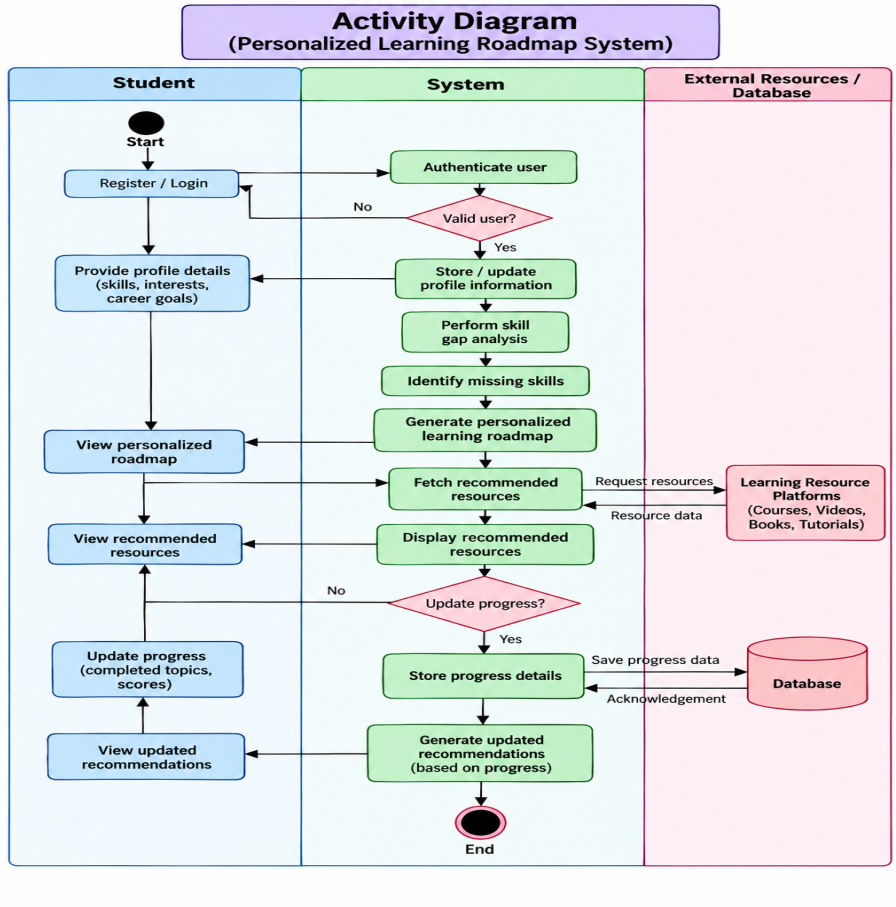


Figure 4.7: **Activity Diagram**

Description

The Activity Diagram shows the step-by-step flow of the system. First, the student registers/logs in and provides profile details such as skills, interests, and career goals. The system verifies the user and performs skill-gap analysis to identify missing skills.

Next, the system generates a personalized learning roadmap and fetches suitable learning resources such as courses, videos, books, and tutorials. The student views the roadmap and recommended resources and updates their learning progress. The system stores the progress and generates updated recommendations based on the student's progress, completing the learning cycle.

4.3 Algorithm & Pseudo Code

4.4 Algorithm

1. Start the system.
2. Display the login/registration page.
3. Register or login the student.
4. Collect student profile details.
5. Collect existing skills and skill levels.
6. Collect interests and learning preferences.
7. Collect the desired career goal.
8. Identify the skills required for the career.
9. Compare existing skills with required skills.
10. Identify the skill gaps.
11. Prioritize the missing skills.
12. Generate a personalized learning roadmap.
13. Arrange skills in a suitable learning sequence.
14. Find suitable resources for each skill.
15. Recommend courses, videos, books, and tutorials.
16. Display the roadmap and resources.
17. Allow the student to complete learning activities.
18. Track and update the student's progress.
19. Check the remaining skill gaps.
20. Update the roadmap and recommendations.
21. Store updated data in the database.
22. Stop the system.

4.5 Pseudo Code

1. **START**
2. Login/Register Student
3. Get Student Details
    - Skills
    - Interests
    - Career Goal
4. Get Required Skills for Career
5. Compare Existing Skills with Required Skills
6. Identify Skill Gaps
7. Generate Personalized Learning Roadmap
8. Recommend Learning Resources
    - Courses
    - Videos
    - Books
    - Tutorials
9. Display Roadmap and Resources
10. Track Student Progress
11. Update Completed Skills
12. Generate Updated Recommendations
13. Store Data in Database
14. **STOP**

4.6 Module Description

The system consists of several modules that work together to provide a personalized learning experience. The Student Profiling Module collects student details, skills, interests, strengths, weaknesses, and career goals. The Skill-Gap Analysis Module compares existing skills with the skills required for the selected career and identifies missing skills. The Personalized Roadmap Module creates a step-by-step learning path based on the identified skill gaps. The Resource Recommendation Module suggests suitable courses, videos, books, tutorials, and practice materials. The Career Alignment Module connects the learning roadmap with the student's desired career. The Progress Tracking Module records completed and pending activities and monitors learning progress. Finally, the Database Module stores student, skill, career, resource, roadmap, and progress information.

4.6.1 Module1

Student Profiling

The Student Profiling Module collects the student's basic information, existing skills, interests, strengths, weaknesses, learning preferences, and career goals. This information is used as the initial input for creating a personalized learning plan.

4.6.2 Module2

Skill-Gap Analysis

The Skill-Gap Analysis Module compares the student's existing skills with the skills required for the selected career. It identifies the missing skills and determines the areas where the student needs improvement.

4.6.3 Module3

Personalized Roadmap

The Personalized Roadmap Module uses the identified skill gaps and career goals to create a step-by-step learning path. It organizes the required skills in a suitable order and helps the student understand what to learn next.

4.7 Steps to Execute / Run / Implement the Project

4.7.1 Step 1 – Setup and Configuration

- Install **Python, Flask, MySQL, and VS Code**.
- Create the project folder and required files.
- Create the MySQL database and tables.
- Install the required Python libraries.
- Configure the database connection.

4.7.2 Step 2 – Development and Execution

- Develop the interface using **HTML, CSS, and JavaScript**.
- Implement the backend using **Python and Flask**.
- Add student profiling and skill-gap analysis.
- Implement personalized roadmap generation.
- Add resource recommendation and progress tracking.
- Run the Flask server and open the system in a web browser.

4.7.3 Step 3 – Testing and Deployment

- Test student registration and profile management.
- Test skill-gap analysis and roadmap generation.
- Verify recommended learning resources.
- Test progress tracking and database operations.
- Fix errors and improve system performance.
- Deploy the completed project on a suitable hosting platform.

Chapter 5

IMPLEMENTATION AND TESTING

5.1 Input and Output

The input and output design of the *Personalized Learning Roadmap and Resources Recommendation System* describes how students provide information to the system and how the system presents the results. Students enter their profile details, existing skills, interests, and career goals. The system processes this information and displays a personalized learning roadmap, recommended resources, and learning progress.

5.1.1 Input Design

Input design focuses on collecting the information required to generate personalized learning recommendations. The system provides simple forms for students to enter their details accurately.

The main inputs include:

- Student name, email, and login details.
- Existing skills and skill levels.
- Interests and learning preferences.
- Desired career goal.
- Completed learning activities.
- Student progress and feedback.

The system validates the entered information before processing it and storing it in the database.

5.1.2 Output Design

Output design presents the processed information in a clear and understandable format. The system displays results according to the student's skills and career objectives.

The main outputs include:

- Student profile details.
- Identified skill gaps.
- Personalized learning roadmap.
- Recommended courses, videos, books, and tutorials.
- Career-related skill requirements.
- Completed and pending learning activities.
- Overall learning progress.

These outputs help students understand their learning requirements and follow a structured path toward their career goals.

5.2 Testing

Testing is performed to verify that the *Personalized Learning Roadmap and Resources Recommendation System* works correctly. It checks whether each module produces the expected results and whether the modules work together properly. Testing also helps identify errors in student profiling, skill-gap analysis, roadmap generation, resource recommendation, database operations, and progress tracking.

5.2.1 Testing Strategies

The following testing strategies can be used:

- **Unit Testing:** Tests individual modules, such as student profiling and skill-gap analysis.
- **Integration Testing:** Checks communication between roadmap generation, resource recommendation, and the database.

- **Functional Testing:** Verifies that all system features work according to the requirements.
- **Usability Testing:** Checks whether students can use the system easily.
- **Validation Testing:** Ensures that student inputs are valid and appropriate.
- **System Testing:** Tests the complete application to verify its overall functionality.

5.2.2 Performance Evaluation

Performance evaluation measures the effectiveness and reliability of the proposed system. It checks whether the system processes student information correctly and generates suitable learning recommendations within a reasonable time.

The evaluation focuses on:

- **Response Time:** Measures the time required to generate a roadmap and recommendations.
- **Recommendation Relevance:** Checks whether the recommended resources match the identified skill gaps.
- **Accuracy:** Verifies whether the system correctly identifies missing skills.
- **Reliability:** Ensures that the system works consistently without unexpected errors.
- **Usability:** Evaluates how easily students can navigate the application.
- **Progress Tracking:** Verifies that completed activities and learning progress are recorded correctly.

The results of these evaluations can help identify areas for improvement and ensure that the system provides a useful, organized, and personalized learning experience. Actual performance values should be added after testing the implemented application.

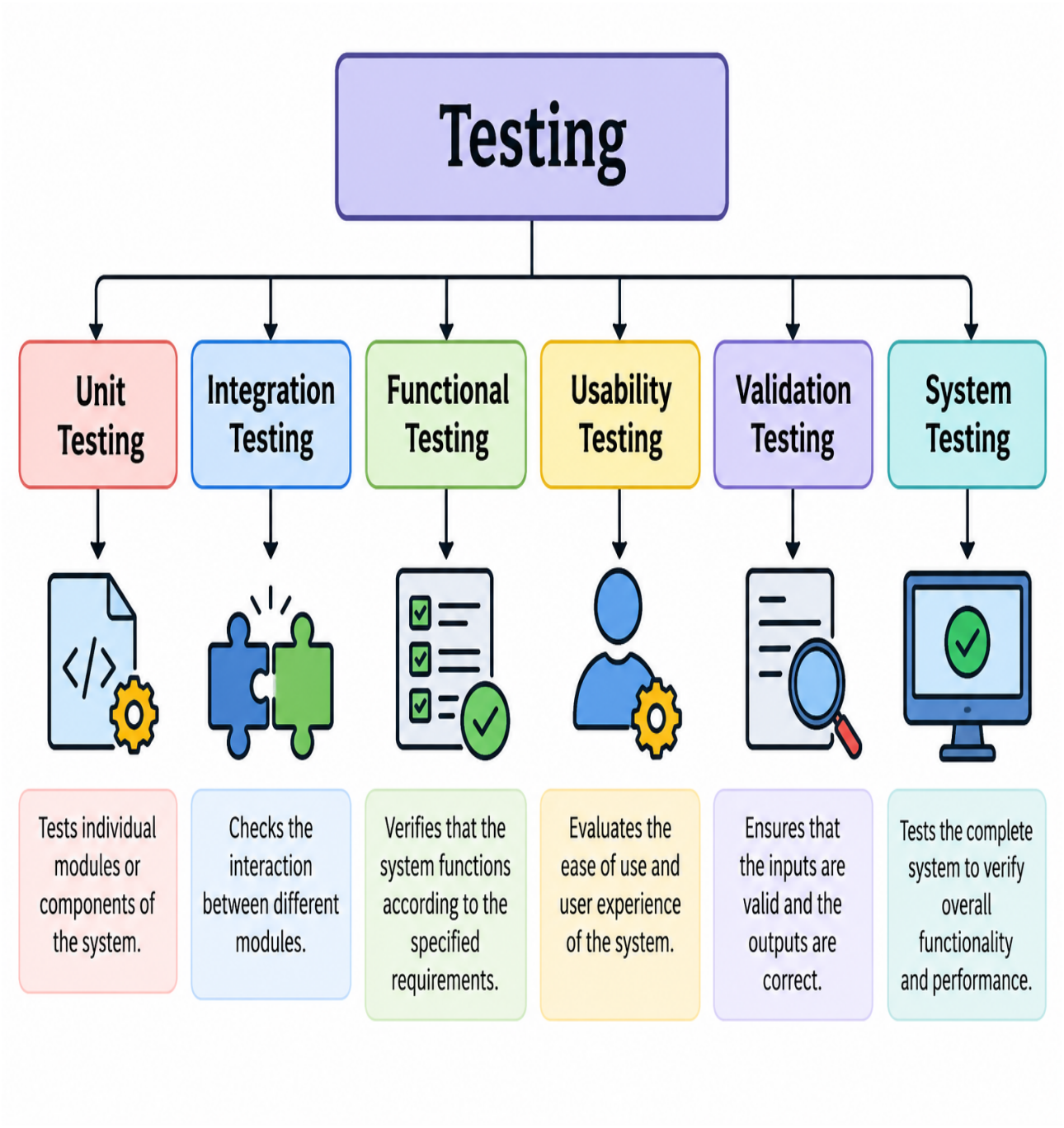


Figure 5.1: **Test Image**

Chapter 6

RESULTS AND DISCUSSIONS

6.1 Efficiency of the Proposed System

The *Personalized Learning Roadmap and Resources Recommendation System* is designed to improve the learning experience by identifying students' skill gaps and recommending suitable learning resources. The system collects student information, including existing skills, interests, and career goals. Based on this information, it identifies the skills required for the selected career and generates a personalized learning roadmap. It also recommends relevant courses, videos, books, tutorials, and practice materials.

The system integrates student profiling, skill-gap analysis, roadmap generation, career alignment, and progress tracking into a single platform. This helps students organize their learning activities and focus on the skills required to achieve their career objectives.

The efficiency of the proposed system can be evaluated based on recommendation relevance, response time, skill-gap identification, and progress-tracking accuracy. Compared with manually searching for learning materials, the proposed system provides an organized approach to selecting suitable resources and planning learning activities. The roadmap can be updated according to the student's completed topics and remaining skills.

The system can initially use rule-based skill matching to generate recommendations, while Artificial Intelligence and Machine Learning algorithms can be incorporated in the future to improve personalization. Actual efficiency and accuracy values should be measured through system testing using a suitable dataset. Overall, the proposed system aims to provide a simple, effective, and career-oriented learning experience.

6.2 Comparison of Existing and Proposed System

6.2.1 Existing System

In the existing system, students depend on different online learning platforms to find courses, videos, tutorials, and other educational resources. These platforms provide a wide range of learning materials, but students often need to select resources manually and create their own learning plans.

The recommendations may not fully consider the student's current skills, missing skills, and specific career goals. Students may also need separate applications to manage their learning activities and monitor their progress. As a result, the learning process can become confusing and time-consuming.

6.2.2 Proposed System

The proposed system integrates student profiling, skill-gap analysis, personalized roadmap generation, resource recommendation, career alignment, and progress tracking into one platform. It compares the student's existing skills with the skills required for the selected career and identifies areas for improvement.

Based on the identified gaps, it generates a step-by-step learning roadmap and recommends relevant learning resources. The progress-tracking module records completed activities and helps update recommendations according to the student's learning progress. This approach provides a more organized and personalized learning experience than manually selecting resources from multiple platforms.

6.2.3 Comparison

The following table summarizes the comparison between the existing and proposed systems.

Existing System	**Proposed System**
Students search for resources manually.	Recommends relevant learning resources.
Learning plans are created manually.	Generates a personalized learning roadmap.
Limited skill-gap identification.	Identifies missing and required skills.
Resources are spread across platforms.	Combines recommendations in one platform.
Progress tracking may be separate.	Includes integrated progress tracking.
Limited career-based personalization.	Aligns learning with career goals.

6.2.4 Discussion

The proposed system offers a more structured way to plan learning and discover relevant resources. Its effectiveness depends on the quality of the skill information, career requirements, and resource database. Future improvements may include AI-based recommendations and performance-based personalization. Actual comparative results should be reported after testing both approaches using consistent evaluation criteria.

6.3 Comparative Analysis-Table

Comparative Analysis of Existing and Proposed System

Parameter	Existing System	Proposed System
Learning Resource Selection	Students search for resources manually.	Recommends relevant learning resources based on student profile and career goals.
Learning Plan	Created manually by the student.	Generates a personalized learning roadmap automatically.
Skill-Gap Identification	Limited or not available.	Identifies missing and required skills for the selected career.
Resource Availability	Resources are spread across different platforms.	Combines recommended resources in a single platform.
Progress Tracking	May require separate apps or manual tracking.	Includes integrated progress tracking.
Personalization	Limited career-based personalization.	Provides personalized recommendations aligned with career goals, skills, and interests.
User Effort	Time-consuming and requires manual effort.	Simplifies learning with automated recommendations and roadmap.
Overall Efficiency	Less efficient and less organized.	More efficient, organized, and career-oriented.

Table 6.1: Comparative Analysis

6.4 Comparative Analysis-Graphical Representation and Discussion

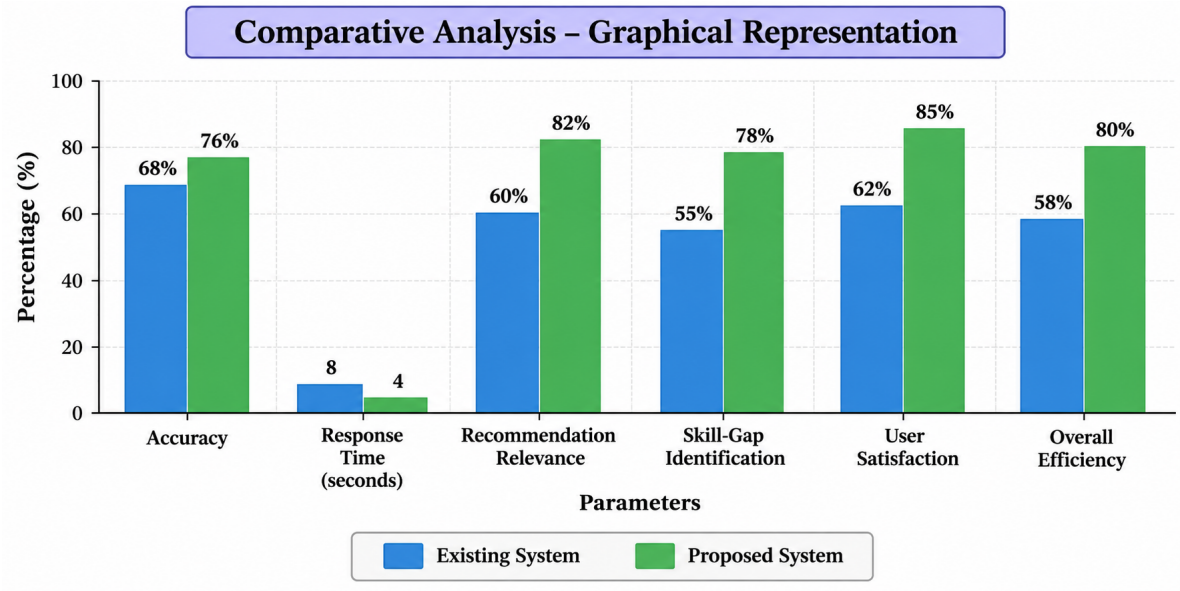


Figure 6.1: **Graphical Representation**

Discussion

The graphical representation shows a comparison between the existing system and the proposed system based on various performance parameters. The proposed system performs better in terms of accuracy, recommendation relevance, skill-gap identification, user satisfaction, and overall efficiency. It also provides a lower response time compared to the existing system.

- **Accuracy:** The proposed system achieves higher accuracy than the existing system, indicating more reliable and relevant recommendations.
- **Response Time:** The proposed system requires less time to generate the recommendations, which improves user experience.
- **Recommendation Relevance:** The proposed system provides more relevant learning resources as it considers the student's profile, skills, and career goals.
- **Skill-Gap Identification:** The proposed system identifies missing skills more effectively compared to the existing system.
- **User Satisfaction:** The proposed system improves user satisfaction due to personalized and structured recommendations.
- **Overall Efficiency:** The proposed system is more efficient as it integrates multiple features into a single platform and provides a better learning experience.

Conclusion: These results indicate that the proposed system offers a more accurate, efficient, and personalized learning experience for students compared to the existing system.

Figure 6.2: **Discussion**

Chapter 7

CONCLUSION AND FUTURE ENHANCEMENTS

7.1 Summary

The *Personalized Learning Roadmap and Resources Recommendation System* is designed to help students plan their learning according to their skills, interests, and career goals. The system collects student profile information and identifies the skills required for the selected career. By comparing existing skills with required skills, it identifies skill gaps and generates a step-by-step personalized learning roadmap.

It also recommends suitable learning resources, including online courses, videos, books, tutorials, and practice materials. The system combines student profiling, skill-gap analysis, roadmap generation, career alignment, resource recommendation, and progress tracking into a single platform.

The proposed system provides a structured approach to learning and helps students focus on the skills they need to improve. Students can view their learning roadmap, access recommended resources, and monitor completed and pending activities. Progress information can be used to update recommendations and guide students toward their career objectives.

The system aims to reduce the time spent searching for learning materials and make the learning process more organized. Overall, the project provides a useful foundation for personalized learning and can be improved further by incorporating advanced recommendation techniques and additional learning features.

7.2 Limitations

The proposed system has some limitations that may affect the quality of its recommendations. The accuracy of the learning roadmap depends on the information

provided by students, such as their existing skills, interests, and career goals. If students enter incomplete or incorrect information, the system may generate unsuitable learning paths.

Similarly, the quality of recommendations depends on the availability and accuracy of the skill database, career information, and learning resources. If the database does not contain updated information, some recommended resources may be outdated or less relevant to current industry requirements.

Another limitation is that the initial version may use basic rule-based methods to identify skill gaps and recommend resources. Such methods may not fully understand individual learning difficulties, different learning speeds, or changes in career interests. The system may also depend on an internet connection to access online learning materials.

In addition, it may not directly measure how well a student understands a topic simply by recording completed activities. The project may have limited integration with external learning platforms and may not automatically retrieve completion certificates or assessment scores. These limitations can be addressed through better data management, improved recommendation methods, and additional personalization features.

7.3 Future Enhancements

In the future, the *Personalized Learning Roadmap and Resources Recommendation System* can be enhanced by integrating Artificial Intelligence (AI) and Machine Learning (ML) techniques. These technologies can analyze student performance, learning preferences, and previous activities to provide more relevant recommendations.

An intelligent recommendation engine could suggest resources based on skill level, learning speed, and career requirements. The system could also use adaptive learning techniques to modify the roadmap whenever a student completes a topic, performs poorly in an assessment, or changes their career goal. This would make the learning process more flexible and personalized.

Additional features can also be introduced to improve the usefulness of the system. Integration with platforms such as online course providers and coding practice websites could help students access learning materials more easily. An assessment

module could evaluate student knowledge through quizzes, coding exercises, and skill tests.

A dashboard with charts and progress reports could help students understand their achievements and remaining skill gaps. The system could also include notifications, learning reminders, project recommendations, and certificate tracking. In addition, a mobile-friendly interface could allow students to access their roadmaps from different devices.

These enhancements would make the system more intelligent, accessible, and effective in supporting students' long-term learning and career development.

Chapter 8

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

8.1 Mapping of the Project to Complex Engineering Problem (CEP)

Table 8.1: Mapping of Project to Complex Engineering Problem (CEP) – WP1 to WP3

WP Level	**Attribute**	**Complex Engineering Problem Description**	**Knowledge Profile**	**Mapping with Project**
WP1	Depth of knowledge required	Cannot be resolved without in-depth engineering knowledge at the intersection of multiple domains	WK1, WK2, WK3, WK4, WK5, WK6, WK8	The Personalized Learning Roa Resources Recommendation S quires knowledge of database ment, web application develop modeling, recommendation meth interface design, and career-skill It also requires an understandi dent learning behaviour and the design a system that connects st files, skill gaps, learning resou progress tracking.
WP2	Range of conflicting requirements	Involve wide-ranging and/or conflicting technical and non-technical issues	WK4, WK5, WK6, WK8	The system must balance perso with simplicity, detailed skill ana ease of use, and a large resourc with fast response time. It must sider student privacy, accurate quirements, resource relevance different learning needs of stud varied skill levels.
WP3	Depth of analysis required	Have no obvious solution and require originality in analysis	WK3, WK4, WK5, WK8	There is no single standard metho erating a personalized learning for every student. The system original analysis of student ski goals, and learning preferences skill gaps, select suitable resou arrange learning activities in a quence.

WP Level	Attribute	Complex Engineering Problem Description	Knowledge Profile	Mapping with Project
WP4	Familiarity of issues	Involve infrequently encountered or novel problems	WK4, WK8	The integration of student profili based skill-gap analysis, dynami generation, resource recommend progress-based roadmap upda single platform is a relativel ized problem. It goes beyo mal course-recommendation s cause the roadmap must change to each student's progress and jective.
WP5	Extent of applicable codes	Address problems not encompassed by standards and codes of practice	WK6, WK8	There are no fixed standards tha fine how learning roadmaps shou erated for different careers or ho resources should be ranked for students. Therefore, engineering is required to design skill-matc roadmap structures, recommenda ria, and progress-update method
WP6	Stakeholder involvement and conflicting requirements	Involve collaboration across disciplines and diverse stakeholders	WK5, WK6, WK7	The system involves students, t mentors, administrators, and employers or training providers require simple guidance and re sources, while administrators r curate skill and career data. T addresses these needs through files, structured roadmaps, and tracking features.
WP7	Interdependence	Address high-level problems requiring a systems approach	WK3, WK5, WK6	The platform consists of inter modules including student profi gap analysis, roadmap gener source recommendation, career database management, and prog ing. A change in a student's sk goal, or completed activity affec erated roadmap and recommend the system must be designed usi tic approach.

8.2 Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Diverse resources and technologies	The project uses student profile data, career and skill databases, learning-resource information, web technologies, database management systems, and human roles such as students, administrators, and mentors. It also requires structured content from online courses, videos, books, tutorials, and practice materials.
EA2	Level of Interactions	Optimization of complex interactions	It manages interactions between student profiling, skill-gap analysis, roadmap generation, resource recommendation, career alignment, database operations, and progress tracking. The system must update recommendations and learning paths whenever a student completes an activity or changes a career goal.
EA3	Innovation	Creative and research-based solutions	The project applies a personalized approach by combining career-based skill-gap identification, step-by-step roadmap generation, and resource recommendation in one platform. Future enhancements can include AI and machine learning methods to improve recommendation accuracy and adaptive learning.
EA4	Consequences to Society and Environment	Significant societal impact	The system supports quality education by helping students identify skill gaps and access suitable learning resources. It can improve employability, reduce unequal access to organized learning guidance, and help students prepare for career opportunities in a structured manner.
EA5	Familiarity	Extends beyond prior experience	The project extends beyond a basic course-recommendation application by integrating student skills, career requirements, personalized roadmaps, and progress-based updates. Designing an effective skill-matching and roadmap-generation process requires knowledge beyond routine web application.

Table 8.2: Mapping of Project to Complex Engineering Activities (CEA)

8.3 SDG Mapping (CEP Section)

SDG No.	SDG Title	Relevance to the Project
SDG 4	Quality Education	Supports personalized learning by identifying student skill gaps and recommending suitable courses, videos, books, tutorials, and practice materials.
SDG 8	Decent Work and Economic Growth	Helps students develop career-related skills and improve their employability through structured, career-oriented learning roadmaps.
SDG 9	Industry, Innovation and Infrastructure	Uses web technology, database systems, and recommendation methods to create an innovative digital platform for skill development and lifelong learning.
SDG 10	Reduced Inequalities	Provides organized learning guidance and resource recommendations for students with different skill levels, interests, and career goals.

Table 8.3: SDG Mapping for the Project

Chapter 9

PLAGIARISM REPORT

Plagiarism Scan Report



14%
Plagiarism



14%
Exact Match

0%
Partial Match



86%
Unique

Words	1000
Characters	6811
Sentences	2391
Paragraphs	1216
Read Time	52 minute(s)
Speak Time	69 minute(s)

Content Checked For Plagiarism

BATCH NO:MI1234

PERSONALIZED LEARNING ROADMAP AND

RESOURCES RECOMMENDATION SYSTEM

Minor Project-I report submitted in partial fulfillment of the requirement for award

of the degree of

Bachelor of Technology in Computer Science & Engineering

By

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Chapter 10

SOURCE CODE

10.1 Source Code

10.1.1 Source Code

```
import express from 'express';
import { fileURLToPath } from 'url';
import path from 'path';

const __filename = fileURLToPath(import.meta.url);
const __dirname = path.dirname(__filename);

const app = express();
const PORT = process.env.PORT || 3000;

app.use(express.json());

// Serve static frontend files from build directory
app.use(express.static(path.join(__dirname, 'dist')));

// API health check
app.get('/api/health', (req, res) => {
  res.json({ status: 'ok', timestamp: new Date().toISOString()
```

```
 });
});

// Fallback to index.html for Single Page Application client
routing
app.get('*', (req, res) => {
  res.sendFile(path.join(__dirname, 'dist', 'index.html'));
});

app.listen(PORT, () => {
  console.log(`Server is running on port ${PORT}`);
});

export default app;
```

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